



A comparison of homogenization methods for 2-D woven composites

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ABSTRACT

Woven composites present several advantages over traditional uni-directional laminates and predicting their mechanical properties is particularly important when designing structures using this kind of materials.

The capability of the asymptotic expansion homogenization method (AEHM) compared to analytical methods based on the mechanics of materials approach in predicting the mechanical properties of two-dimensional woven composites will be evaluated against experimental and numerical results for different fabric architectures and material constituents. In particular the analytical methods that will be taken in to consideration are the parallel–parallel, parallel–series, series–parallel and series–series methods.

The results show that the AEHM gives good predictions compared to experimental results and compares well with the parallel–parallel method, which is the most reliable of the analytical methods considered. Moreover, the AEHM is able to provide localization of stresses and strains as part of the methodology, feature that is not provided by the analytical methods.

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1. Introduction

The use of composite materials in industrial applications has been increasing over the past years thanks to their very high specific strength, high specific stiffness, fatigue characteristics and moderate cost.

Laminar composite materials can support very high loads in the direction of the fibers, but only low loads in the translaminar direction, as all the load is supported only by the matrix. Moreover, laminar composite materials have a poor resistance to the damage caused by the impact of a foreign object on their outer surface, because they dissipate very little impact strain energy (such as plastic yielding in ductile metals) during impact loading.

Composites with woven fiber reinforcements have proven to have superior impact energy absorption capabilities [1] and to have very good in-plane properties in mutually orthogonal directions as well as more balanced properties [2] compared to laminates with uni-directional (UD) fibers.

The mechanical properties of the woven composites will depend on the weave architecture and materials used. For this reason the prediction of the mechanical properties for the woven composites assumes a critical role when designing the laminate.

Following the work of Ishikawa and Chou [3,4], many methods have been proposed to determine the mechanical properties of

two-dimensional (2-D) woven composites. These methods can be broadly categorized in analytical methods based on the mechanics of materials approach (e.g. [2,5–7]), methods based on the finite element analysis (e.g. [8–10]) and methods based on the asymptotic expansion homogenization method (AEHM) approach (e.g. [11–15]).

The present paper aims to compare the AEHM with the analytical methods in order to assess its capability in predicting the mechanical properties for different fabric architectures and material constituents.

Section 2 summarizes the theory behind the AEHM and the analytical methods, while Section 3 presents and discusses the numerical results. In Section 4 some conclusions are drawn on the effectiveness of these methods.

2. Methods

An orthogonal 2-D woven fabric consists of two sets of yarns (or strands) interlaced at 90° to each other. In the following the terms yarns and strands will be used interchangeably. The lengthwise yarn is called *warp*, while the crosswise yarn is called *fill* (often also *weft*). The weave patterns repeat every certain number of warp yarns and fill yarns. This basic area, which is the smallest periodic region of the woven composite that repeats itself in the lamina, is a complete *representative volume element* (RVE) of the weave, and is sufficient to estimate its mechanical properties. Note that here the RVE coincides with the *repeating unit cell* (RUC) typical of periodic structures. Refer to Pindera et al. [16] for the definitions of RVE and

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RUC and a discussion on the most appropriate uses of the two expressions. In the following the term RVE will be used to conform with previous literature.

There are various weave styles of orthogonal 2-D woven composites based on the way the yarns are interlaced. The different weave patterns are characterized by the geometrical parameters n_g^f and n_g^w , which represent the n th yarn with which an orthogonal yarn is interlaced. For example a warp yarn is interlaced with every n_g^f fill yarn and a fill yarn is interlaced with every n_g^w warp yarns. Fig. 1 [17, p. 71] gives some examples of RVE for some typical 2-D weaves. If the yarns have the same size and same undulation length and $n_g^f = n_g^w = n_g$ then the fabric is called *balanced*; otherwise it is called *unbalanced*. Moreover, if the two sets of yarns are of different materials, then the fabric is called a *hybrid weave* [7]. In general, the RVE is characterized by two parameters: (1) the fabric type n_g (see Fig. 1) and, for the satin fabrics, (2) the satin weave type n_{dec} , which accounts for the position of the interlaced region in the RVE. In the following, $n_{dec} = 1$ is assumed.

Once the RVE of the 2-D woven lamina has been identified, this area is sub-divided into smaller regions, called *elements*, based on the undulation of the warp and fill yarn, and position of the yarn in the element [18]. Three different types of elements (in the material coordinate system of the fibers) are generally present in a woven lamina: (1) those composed of both matrix and reinforcement fibers in the fill direction; (2) those composed of both matrix and reinforcement fibers in the warp direction; and finally (3) those composed only of matrix. Also there is the possibility of a presence of a gap g_f between the fill yarns and a gap g_w between the warp yarns. In Section 3, the presence of gaps will be considered.

With regards to notation, the *shorthand* (or *contracted*) notation [19, p. 17] for Cartesian coordinates is used for stress σ and strain ϵ . The coordinate axes are an x – y – z system or a 1–2–3 system. The x – y – z system is called the *global* system, while the 1–2–3 system is the *material* or *on-axis* system. All rotations of coordinate axes are assumed to be about the z -axis, so z is coincident with the 3-direction, which is consistent with the assumption that individual laminae are modeled as orthotropic materials. By convention, axis 1 is

the direction parallel to the reinforcement fibers, axis 2 is the in-plane direction perpendicular to fibers and axis 3 is the out-of-plane direction perpendicular to fibers.

For the matrix the hypothesis of *isotropic* material is assumed, i.e. all planes are planes of material symmetry and are isotropic. The reinforcement fibers can be assumed to be *isotropic* materials or *transversely isotropic* materials, i.e. the material has an axis of symmetry, e.g. the 1-axis, and any two material fibers, having symmetrical position to the axis of symmetry, have the same stiffness. In this case, the 2-axis and 3-axis can be in any direction (but perpendicular to each other) without altering the value of the compliance and the x – y plane is an isotropic plane.

The mechanical properties of the elements composed of two constituents will be derived using micro-mechanical models. Micro-mechanical models are generally used to predict the mechanical properties of uni-directional (UD) fiber-reinforced laminae, so the underlying assumption here is that the yarns (strands) have the same mechanical behavior as in UD laminae.

Various micro-mechanics models can be used to calculate the equivalent mechanical properties of the strands. See e.g. [20] for a review of existing micro-mechanics models. The *rules of mixtures* model [19] gives very good estimates in the longitudinal direction, but is inaccurate in the transverse direction. Other micro-mechanics models give a better performance in the transverse direction, but at the expense of an increased computational cost. Lamers [21] suggests, as a compromise, employing the rules of mixtures only in the longitudinal direction, and a more precise (but more complex) method only in the transverse direction. In particular here (1) the models termed Chamis [22] and Hashin and Rosen or *Composite Cylinder Assemblage* (CCA) for *anisotropic constituents* [23] were used for the elements composed of matrix and transversely isotropic reinforcement fibers; (2) the model termed Hashin and Rosen or CCA model for *isotropic constituents* [24] was used for the elements composed of matrix and isotropic reinforcement fibers.

In the micro-mechanics models one must consider the fiber volume fraction in the (warp and fill) yarns. As discussed by [25] and [21], the fiber volume fractions in the warp and fill yarns $V_f^{w,f}$ are related to the total fiber volume fraction in the composite V_f by the following equation:

$$V_f^{w,f} = \frac{\Omega}{\Omega^{w,f}} V_f \quad (2.1)$$

where Ω is the total volume of the composite and $\Omega^{w,f}$ is the volume of the warp and fill yarns. V_f is determined by the packing structure, and for square packing is [21,18]

$$V_f = \frac{\frac{1}{4}\pi d^2}{d^2} = \frac{\pi}{4} \approx 0.785 \quad (2.2)$$

where d is the diameter of the fibers. V_f for the hexagonal packing is [21,18]

$$V_f = \frac{\frac{1}{8}\pi d^2}{\frac{1}{4}\sqrt{3}d^2} = \frac{\pi}{2\sqrt{3}} \approx 0.907 \quad (2.3)$$

In order to calculate the mechanical properties of the woven lamina in the global coordinates, the geometrical characteristics and the orientation angles of the reinforcement fibers in the fill and warp strands must be taken into account. As discussed by various authors [25–27,21] the average mechanical properties for an element in the woven lamina are calculated considering the fill and warp yarn mid-plane functions $m_f(x)$, $m_w(y)$ and the fill and warp yarn cross-sectional shape functions $s_f(y)$, $s_w(x)$. The shape functions delimit regions in the woven composite with pure matrix material and regions with impregnated yarns. The height position

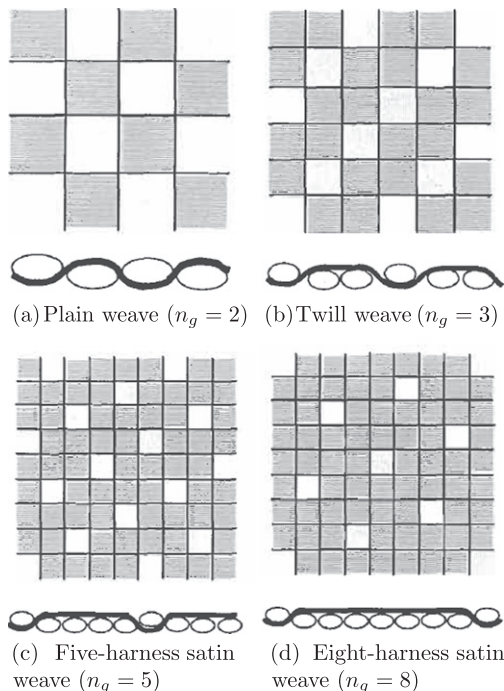


Fig. 1. Representative volume elements of typical 2-D woven composites (reprinted with permission from [17, p. 71]).

of the yarn surface h in each element is a sum of the yarn mid-plane shape functions and the cross-sectional yarn shape functions for that element.

$$\begin{aligned} h_{F_T}(x, y) &= m_F(x) + s_F(y) \\ h_{F_B}(x, y) &= m_F(x) - s_F(y) \\ h_{W_T}(x, y) &= m_W(y) + s_W(x) \\ h_{W_B}(x, y) &= m_W(y) - s_W(x) \end{aligned} \quad (2.4)$$

where $h_{F_T}(x, y)$, and $h_{F_B}(x, y)$ define the fill top and bottom surface functions, while $h_{W_T}(x, y)$ and $h_{W_B}(x, y)$ define the warp top and bottom surface functions. In defining the mid-plane and surface function, one might also account for the presence of gaps between the fill and warp strands, as will be discussed in Section 3. The fiber undulation for the fill and warp strands are a function of the yarn mid-plane shape functions as follows:

$$\theta_F(x) = \arctan \frac{dm_F(x)}{dx} \quad (2.5)$$

$$\theta_W(y) = \arctan \frac{dm_W(y)}{dy} \quad (2.6)$$

The stress transformation referred to the x - y - z coordinate system (global system) to those in the 1-2-3 coordinate system (material system) is [27]

$$\bar{\mathbf{Q}} = \mathbf{T}^{-1} \mathbf{Q} \mathbf{R} \mathbf{T} \mathbf{R}^{-1} \quad (2.7)$$

where $\mathbf{Q}$ is the transformed stiffness matrix in the material coordinate system; $\bar{\mathbf{Q}}$ is the transformed stiffness matrix in the global coordinate system; $\mathbf{T}$ is the stress transformation matrix; $\mathbf{R}$ is the Reuter matrix.

2.1. Asymptotic expansion homogenization method

As the woven composite has a periodic structure based on the RVE, then it is possible to apply the asymptotic expansion homogenization method (AEHM). The mathematical basis of this method were placed by [28,29]. Recent reviews on the AEHM applied to composites are [30–32].

The AEHM is characterized by analyzing the material at several scales of analysis. For the woven material it is possible to identify in general three *scales* (or *levels*): (1) the *micro* scale for the constituents (i.e. fibers and matrix); (2) the *meso* scale for the strands and (3) the *macro* scale for the RVE. Some authors [21] refer to macro scale as *mini* scale reserving the term macro scale to the laminate. Here only the two scale analysis will be discussed, specifically the meso (or local) and macro (or global) scales, assuming that the mechanical properties of the strands at the meso scale have been homogenized using micro-mechanics models, as discussed previously in this section. Refer to [14] for the case of three scales analysis applied to woven composites.

The AEHM assumes a small parameter, often called *scale parameter* [15] and referred to here as ϵ , which is the ratio between the two length scales being considered. The two sets of parameters for the macro scale $\mathbf{x}$ and meso scale $\mathbf{y}$ are then related to by the following equation:

$$\mathbf{x} = \frac{\mathbf{y}}{\epsilon} \quad (2.8)$$

The conventional equations of elasticity for a heterogeneous body are given by [30]

$$\frac{\partial \sigma_{ij}^\epsilon(\mathbf{x})}{\partial x_j^\epsilon} + f_i(\mathbf{x}) = 0 \quad (2.9)$$

$$\sigma_{ij}^\epsilon(\mathbf{x}) = a_{ijkl}^\epsilon(\mathbf{x}) \epsilon_{kl}(\mathbf{u}^\epsilon(\mathbf{x})) \quad (2.10)$$

$$\epsilon_{ij}(\mathbf{u}^\epsilon(\mathbf{x})) = \frac{1}{2} \left(\frac{\partial u_i^\epsilon(\mathbf{x})}{\partial x_j^\epsilon} + \frac{\partial u_j^\epsilon(\mathbf{x})}{\partial x_i^\epsilon} \right) \quad (2.11)$$

where (2.9) is the *balance equations*, (2.10) is the *constitutive equations* and (2.11) is the *strain definition*. The super-script ϵ is used to indicate that the variables of the problem depend on the true, high-resolution behavior of the material under consideration. The boundary and discontinuity conditions are the following

$$\sigma_{ij}^\epsilon(\mathbf{x}) n_j = 0 \text{ on } \partial_1 \Omega \quad (2.12)$$

$$u_i^\epsilon(\mathbf{x}) = 0 \text{ on } \partial_2 \Omega \quad (2.13)$$

$$[u_i^\epsilon(\mathbf{x})] = 0 \text{ on } S_j \quad (2.14)$$

$$[\sigma_{ij}^\epsilon(\mathbf{x}) n_j] = 0 \text{ on } S_j \quad (2.15)$$

Square parentheses denote the jump in the enclosed value. $\mathbf{u}$ is the displacement vector, $\epsilon_{ij}(\mathbf{u}(\mathbf{x}))$ is the strain tensor, $\sigma_{ij}(\mathbf{x})$ is the stress tensor, $a_{ijkl}(\mathbf{x})$ is the elasticity tensor and $f_i(\mathbf{x})$ are the volume forces.

The physical quantities describing the mechanical behavior of the body are approximated with an asymptotic series representation in ϵ given by

$$\mathbf{u}^\epsilon(\mathbf{x}) = \mathbf{u}^{(0)}(\mathbf{x}) + \epsilon \mathbf{u}^{(1)}(\mathbf{x}, \mathbf{y}) \quad (2.16)$$

$$\boldsymbol{\sigma}^\epsilon(\mathbf{x}) = \boldsymbol{\sigma}^{(0)}(\mathbf{x}) + \epsilon \boldsymbol{\sigma}^{(1)}(\mathbf{x}, \mathbf{y}) \quad (2.17)$$

$$\boldsymbol{\epsilon}^\epsilon(\mathbf{x}) = \boldsymbol{\epsilon}^{(0)}(\mathbf{x}) + \epsilon \boldsymbol{\epsilon}^{(1)}(\mathbf{x}, \mathbf{y}) \quad (2.18)$$

Here the expansions have been limited to a two scale approximation. The quantities $\mathbf{u}^{(k)}$, $\boldsymbol{\sigma}^{(k)}$ and $\boldsymbol{\epsilon}^{(k)}$ take the same values on opposite sides of the RVE (i.e. *periodic boundary conditions*). Moreover, the introduction of the second coordinate system $\mathbf{y}$ means that the derivatives originally with respect to x_i^ϵ must now make use of the chain rule of differentiation given by [30]

$$\frac{\partial}{\partial x_i^\epsilon} f = \left(\frac{\partial}{\partial x_i} + \frac{1}{\epsilon} \frac{\partial}{\partial y_i} \right) f = \frac{\partial f}{\partial x_i} + \frac{1}{\epsilon} \frac{\partial f}{\partial y_i} \quad (2.19)$$

Substituting the approximations in Eqs. (2.16)–(2.18) into the strain definition Eq. (2.11) and substituting the resulting strain into the constitutive Eq. (2.10) gives an equation dependent on the powers of ϵ . Each coefficient of ϵ is identically zero to ensure that the asymptotic series approximation is valid as $\epsilon \rightarrow 0$ [13]. When the first two coefficients are set equal to zero, the equilibrium equations can be reformulated as the following two hierarchical equations

$$\frac{\partial \sigma_{ij}^{(0)}}{\partial y_j}(\mathbf{x}, \mathbf{y}) = 0 \quad (2.20)$$

$$\frac{\partial \sigma_{ij}^{(0)}}{\partial x_j}(\mathbf{x}, \mathbf{y}) + \frac{\partial \sigma_{ij}^{(1)}}{\partial y_j}(\mathbf{x}, \mathbf{y}) + f_i(\mathbf{x}) = 0 \quad (2.21)$$

while the constitutive equations takes the form

$$\sigma_{ij}^\epsilon(\mathbf{x}, \mathbf{y}) = a_{ijkl}^\epsilon(\mathbf{y}) \left(\frac{\partial \epsilon_{kl}}{\partial x_l}(\mathbf{u}^{(0)}) + \frac{\partial \epsilon_{kl}}{\partial y_l}(\mathbf{u}^{(1)}) \right) \quad (2.22)$$

Eq. (2.21) is the so-called *micro-equation* as it relates the micro perturbation term $\mathbf{u}^{(1)}$ to the zeroth order term $\mathbf{u}^{(0)}$.

The solution of the perturbative displacement $\mathbf{u}^{(1)}(\mathbf{x}, \mathbf{y})$ is normally assumed [28,29] to take the following form with separated variables

$$\mathbf{u}_i(\mathbf{x}, \mathbf{y}) = \frac{\partial \epsilon_p}{\partial x_q}(\mathbf{u}^{(0)}(\mathbf{x})) \chi_i^{pq}(\mathbf{y}) + C_i(\mathbf{x}) \quad (2.23)$$

where the functions $\chi_i^{pq}(\mathbf{y})$ are called the *homogenization functions* [30] for the displacement or *characteristic displacements* [31] or *elastic correctors* [15], and $C_i(\mathbf{x})$ is a constant of integration.

Given (2.23), the first order component of the equations of equilibrium (2.21) can be formulated as the following boundary value problem

find $\chi_i^{pq} \in V_Y$ such that : $\forall v_i \in V_Y$

$$\int_Y a_{ijkl}(\mathbf{y}) \left(\delta_{ip} \delta_{jq} + \frac{\partial \chi_i^{pq}}{\partial y_j}(\mathbf{y}) \right) \frac{\partial v_k}{\partial y_l}(\mathbf{y}) d\Omega = 0 \quad (2.24)$$

where V_Y is the set of kinematically admissible functions that contains the functions with equal values on the opposite side of the RVE. The six vectors χ^{pq} depend only on the geometry of the RVE and on the jumps of material coefficients across S_j . Functions $v_i(\mathbf{y})$ are test functions defined on the RVE.

In (2.16) the dependence on $\mathbf{x}$ alone occurs only on the first term. The functions depending only on $\mathbf{x}$ define the *global* behavior of the structure. The global behavior of stresses the following mean values are defined over the RVE

$$\bar{\sigma}_{ij}^{(0)}(\mathbf{x}) = \frac{1}{|Y|} \int_Y \sigma_{ij}^{(0)}(\mathbf{x}, \mathbf{y}) dY \quad (2.25)$$

which gives rise to the following effective constitutive equation

$$\bar{\sigma}_{ij}^{(0)}(\mathbf{x}) = a_{ijkl}^h \epsilon_{kl}^{(0)}(\mathbf{x}) \quad (2.26)$$

where

$$a_{ijkl}^h = \frac{1}{|Y|} \int_Y a_{ijpq}(\mathbf{y}) \left(\delta_{kp} \delta_{lq} + \frac{\partial \chi_k^{pq}}{\partial y_l}(\mathbf{y}) \right) dY \quad (2.27)$$

The woven composite can now be studied as a homogeneous material with the effective properties given by (2.27).

2.2. Mechanics of materials based approach

In the mechanics of materials based approach the RVE is sub-divided into smaller elements, which in turn may be divided into even smaller regions, called *sub-elements*, based again on the undulation of the warp and fill yarn and the position of the yarns in the element [18]. The mechanical properties of the RVE of the 2-D woven lamina are calculated from those of the elements using averaging approach [27,5,21]. There are four different methods for carrying out this operation, depending on how the stiffness and compliance matrices are averaged across the RVE.

In the *parallel-parallel (PP) scheme* [21,26,27] one assumes iso-strain conditions and averages across the stiffness matrices of the elements as follows:

$$\left(\tilde{A}_{ij}^{el,PP}, \tilde{B}_{ij}^{el,PP}, \tilde{D}_{ij}^{el,PP} \right) = \frac{1}{l_x} \frac{1}{l_y} \int_0^{l_x} \int_0^{l_y} \left(A_{ij}^{el}(x, y), B_{ij}^{el}(x, y), D_{ij}^{el}(x, y) \right) dx dy \quad (2.28)$$

where l_x is the length of the RVE across the x -direction, while l_y is the length of the RVE across the y -direction. The values $\tilde{A}_{ij}^{el,PP}, \tilde{B}_{ij}^{el,PP}, \tilde{D}_{ij}^{el,PP}$ represent an upper boundary for the stiffness of the RVE. These values are used to calculate the compliance matrices $\tilde{a}_{ij}^{el,PP}, \tilde{b}_{ij}^{el,PP}, \tilde{d}_{ij}^{el,PP}$ which in turn are used to calculate the elastic engineering constants of the woven lamina.

In the *parallel-series (PS) scheme* [5] one first assumes iso-strain conditions and averages the stiffness matrices along the x -axis determining the average stiffness matrices along the y -axis.

$$\left(\tilde{A}_{ij}^{el,P}(y), \tilde{B}_{ij}^{el,P}(y), \tilde{D}_{ij}^{el,P}(y) \right) = \frac{1}{l_x} \int_0^{l_x} \left(A_{ij}^{el}(x, y), B_{ij}^{el}(x, y), D_{ij}^{el}(x, y) \right) dx \quad (2.29)$$

The average in-plane stiffness matrices along the y -axis $\tilde{A}_{ij}^{el,P}(y), \tilde{B}_{ij}^{el,P}(y), \tilde{D}_{ij}^{el,P}(y)$ are then used to calculate the compliance matrices $\tilde{a}_{ij}^{el,P}(y), \tilde{b}_{ij}^{el,P}(y), \tilde{d}_{ij}^{el,P}(y)$ along the y -axis, which, assuming now iso-stress conditions, are then integrated along the y -axis.

$$\left(\tilde{a}_{ij}^{el,PS}, \tilde{b}_{ij}^{el,PS}, \tilde{d}_{ij}^{el,PS} \right) = \frac{1}{l_y} \int_0^{l_y} \left(\tilde{a}_{ij}^{el,P}(y), \tilde{b}_{ij}^{el,P}(y), \tilde{d}_{ij}^{el,P}(y) \right) dy \quad (2.30)$$

In the *series-parallel (SP) method* [5] one first assumes iso-stress conditions and averages the compliance matrices along the y -axis determining the average compliance matrices along the x -axis.

$$\left(\tilde{a}_{ij}^{el,S}(x), \tilde{b}_{ij}^{el,S}(x), \tilde{d}_{ij}^{el,S}(x) \right) = \frac{1}{l_y} \int_0^{l_y} \left(a_{ij}^{el}(x, y), b_{ij}^{el}(x, y), d_{ij}^{el}(x, y) \right) dy \quad (2.31)$$

The average compliance matrices along the x -axis $\tilde{a}_{ij}^{el,S}(x), \tilde{b}_{ij}^{el,S}(x), \tilde{d}_{ij}^{el,S}(x)$ are then used to calculate the average stiffness matrices $\tilde{A}_{ij}^{el,S}(x), \tilde{B}_{ij}^{el,S}(x), \tilde{D}_{ij}^{el,S}(x)$ along the x -axis, which, assuming now iso-strain conditions, are then integrated along the x -axis.

$$\left(\tilde{A}_{ij}^{el,SP}, \tilde{B}_{ij}^{el,SP}, \tilde{D}_{ij}^{el,SP} \right) = \frac{1}{l_x} \int_0^{l_x} \left(\tilde{A}_{ij}^{el,S}(x), \tilde{B}_{ij}^{el,S}(x), \tilde{D}_{ij}^{el,S}(x) \right) dx \quad (2.32)$$

In the *series-series (SS) scheme* [21] one assumes iso-stress conditions and then averages across the compliance matrices of the elements as follows:

$$\left(\tilde{a}_{ij}^{el,SS}, \tilde{b}_{ij}^{el,SS}, \tilde{d}_{ij}^{el,SS} \right) = \frac{1}{l_x} \frac{1}{l_y} \int_0^{l_x} \int_0^{l_y} \left(a_{ij}^{el}(x, y), b_{ij}^{el}(x, y), d_{ij}^{el}(x, y) \right) dx dy \quad (2.33)$$

The values $\tilde{a}_{ij}^{el,SS}, \tilde{b}_{ij}^{el,SS}, \tilde{d}_{ij}^{el,SS}$ are used to calculate the elastic engineering constants of the woven lamina, and to obtain the average stiffness matrices $\tilde{A}_{ij}^{el,SS}(x), \tilde{B}_{ij}^{el,SS}(x), \tilde{D}_{ij}^{el,SS}(x)$, which represent a lower boundary for the stiffness of the RVE.

The effective properties of the woven lamina can be calculated [33] (dropping the super-scripts) as follows:

$$\begin{aligned} \tilde{E}_{11} &= \frac{1}{h_t \tilde{a}_{11}} & \tilde{E}_{22} &= \frac{1}{h_t \tilde{a}_{22}} & \tilde{E}_{33} &= \frac{1}{h_t \tilde{a}_{33}} \\ \tilde{G}_{23} &= \frac{1}{h_t \tilde{a}_{44}} & \tilde{G}_{13} &= \frac{1}{h_t \tilde{a}_{55}} & \tilde{G}_{12} &= \frac{1}{h_t \tilde{a}_{66}} \\ \tilde{\nu}_{12} &= -\frac{\tilde{a}_{12}}{\tilde{a}_{11}} & \tilde{\nu}_{13} &= -\frac{\tilde{a}_{13}}{\tilde{a}_{11}} & \tilde{\nu}_{23} &= -\frac{\tilde{a}_{23}}{\tilde{a}_{22}} \\ \tilde{\nu}_{21} &= -\frac{\tilde{a}_{21}}{\tilde{a}_{22}} & \tilde{\nu}_{31} &= -\frac{\tilde{a}_{31}}{\tilde{a}_{33}} & \tilde{\nu}_{32} &= -\frac{\tilde{a}_{32}}{\tilde{a}_{33}} \end{aligned} \quad (2.34)$$

where h_t is the thickness of the lamina.

3. Results

In this section the equivalent mechanical properties predicted by the AEHM and analytical model were compared against experimental results found in literature and those from other theoretical models for the same examples. The results for all four averaging schemes (PP, PS, SP and SS), discussed in Section 2, are reported in the following sub-sections. The models were developed in MATLAB.

Three types of weave fabrics were taken into account: plain weave, five-harness (5H) satin weave and eight-harness (8H) satin weave fabrics. Moreover examples of unbalanced weaves and weaves with the presence of gaps between the yarns were also considered.

It should be noted that while the predicted results are for laminae, the experimental results are normally for laminate. It also should be noted, as pointed out by [6], that not all the mechanical properties are reported in literature because of the difficulties to

measure for example the out-of-plane stiffnesses E_3 and G_{13} , measurement that would require very thick test samples which are very difficult to manufacture. Furthermore, Naik and Ganesh [25] studied the sensitivity of the predicted parameters with respect to the height to width ratio a/h of the yarns for a balanced weave. They found that, in the case of transversely isotropic fibers, increasing the ratio a/h corresponds to a decrease in E_1 and an increase in ν_{12} , while G_{12} remains constant; in the case of isotropic fibers, increasing the ratio a/h , all three parameters E_1 , E_{12} , G_{12} remain practically constant.

The geometrical parameters in the analyzed examples were normally the average values, within a certain range, determined from measurements using an optical microscope.

If the values of the fiber volume fraction of fill and warp yarns were available then these values were used in the calculations. Otherwise, the values of the fiber volume fraction of fill and warp yarns were estimated from the overall fiber volume fraction in the composite, using Eq. (2.1). Akkerman [6] notes that there is usually a scattering of 1–2% in the fibre volume fractions during the manufactures of the test samples due to the moulding process, scattering which impacts the predicted results using analytical models versus the measured ones.

When not directly available, the composite ply thickness was calculated using the following equation [21]

$$h_t = \frac{\rho_w}{V_f^c \rho_f} \quad (3.1)$$

where V_f^c is the composite fiber volume fraction, ρ_f is the fiber density and ρ_w is the areal density of the weave (supplied by the manufacturer). When not directly available, the fill yarn width a_f and the warp yarn width a_w were calculated from the values of the fill yarn counts c_f and warp yarn counts c_w

$$a_f = \frac{1}{c_f} \quad a_w = \frac{1}{c_w} \quad (3.2)$$

As was discussed in Section 2, the following micro-mechanics models were used to calculate the equivalent mechanical properties of the yarns: (1) the Chamis model [22] and the CCA model for anisotropic constituents [23] for sub-elements composed of matrix and transversely isotropic reinforcement fibers and (2) the CCA model for isotropic constituents [24] for sub-elements composed of matrix and isotropic reinforcement fibers. For anisotropic constituents all calculations were repeated for both available models in order to check for differences in the results. As only a very small difference was found, only the results using Chamis model are reported.

The sinusoidal functions developed by Scida et al. [26] were used for mid-plane and cross-sectional shape functions. These

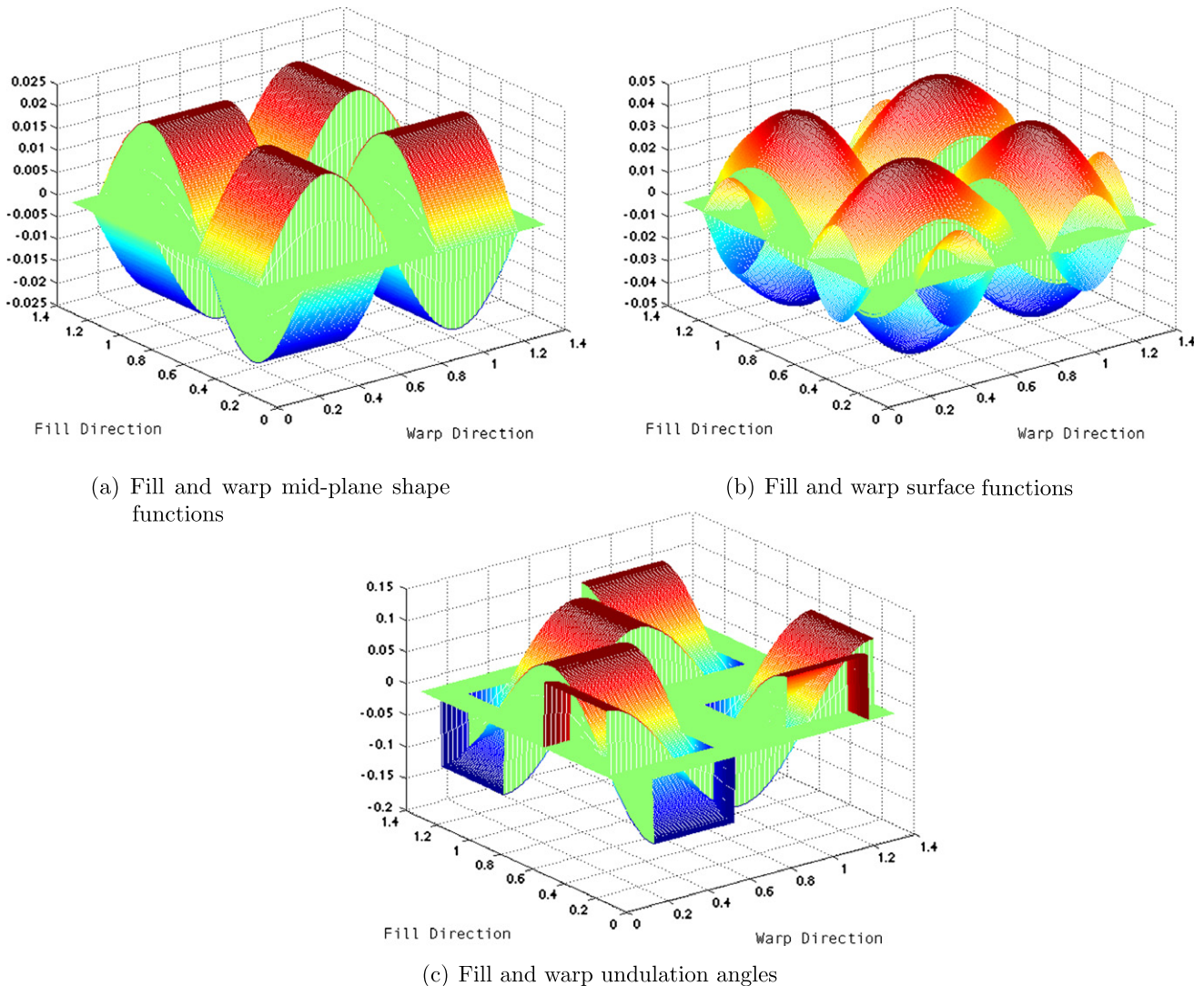


Fig. 2. RVE of plain weave fabric with gaps.

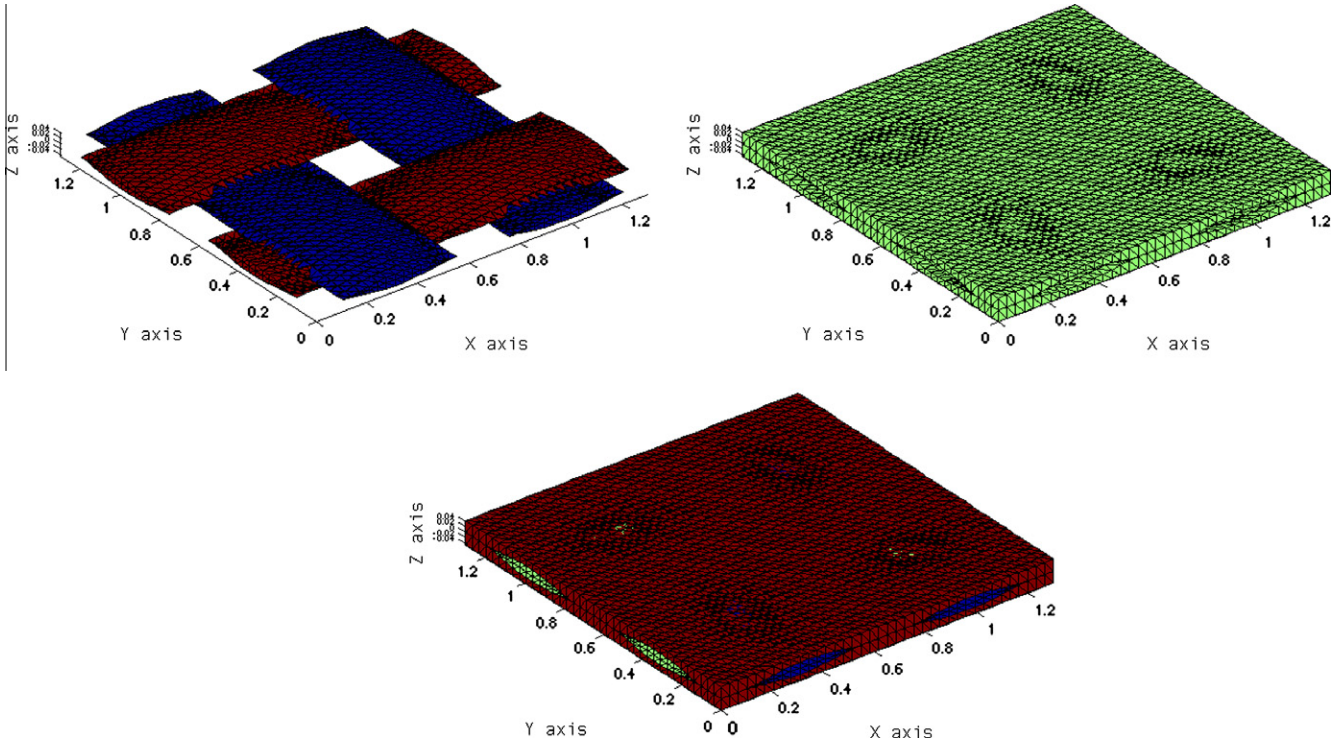


Fig. 3. Tetrahedral mesh for a RVE of plain weave fabric with gaps: strands (a), matrix (b) and complete RVE (c).

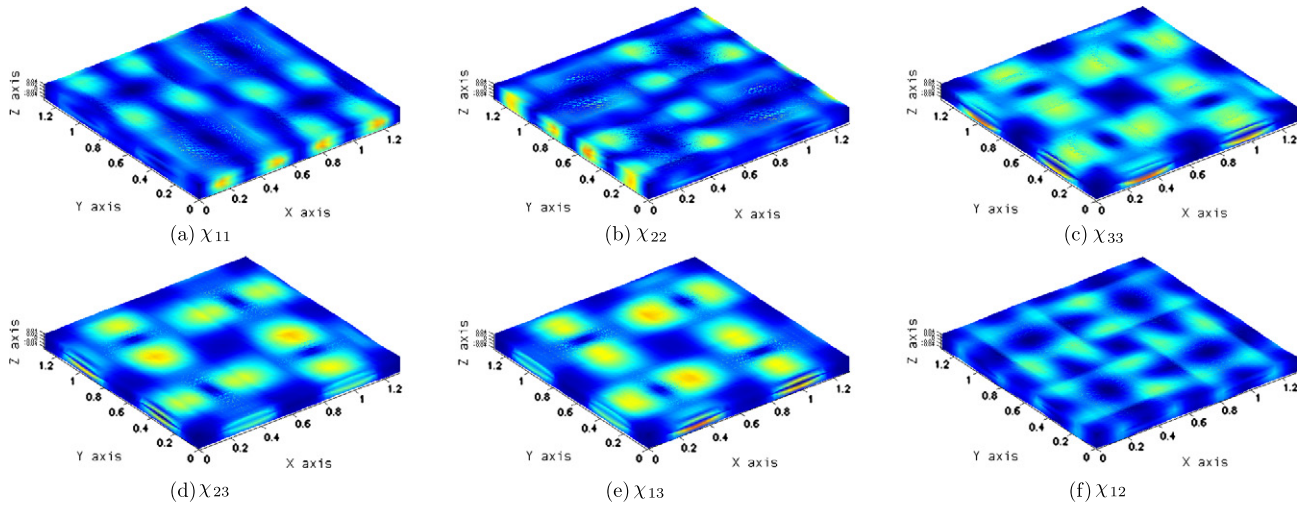


Fig. 4. Microcorrectors for a RVE of plain weave fabric with gaps.

functions were extended in order to account for the presence of a gaps in the fill yarns (g_f) and in the warp yarns (g_w). Although more complex shape functions are available in literature (e.g. [2,7]), the ones in [26] provide satisfactory results, as will be shown in the following. Fig. 2 displays, respectively, the fill and warp mid-plane shape functions, the top and bottom surface functions and the undulation angles for a RVE of plain weave fabric with gaps between the yarns.

In general, as discussed in [34,35], the asymptotic expansion homogenization can be implemented using two different approaches: the *displacement method* (e.g. [11,12]) and the *force method* (e.g. [13,15]). The force method in an ad-hoc code written in MATLAB was used. As explained in [15,30,31] this type of approach could have also been easily implemented in a standard finite element code.

For a hexahedral RVE with $y_1 \in [0, y_1^0]$, $y_2 \in [0, y_2^0]$ and $y_3 \in [0, y_3^0]$ the following periodicity conditions [15,31] were imposed on the surface boundaries of the RVE

$$\begin{aligned}\chi_i^{jk}(0, y_2, y_3) &= \chi_i^{jk}(y_1^0, y_2, y_3), \\ \chi_i^{jk}(y_1, 0, y_3) &= \chi_i^{jk}(y_1, y_2^0, y_3), \\ \chi_i^{jk}(y_3, y_2, 0) &= \chi_i^{jk}(y_1, y_2, y_3^0)\end{aligned}\quad (3.3)$$

An in-house automatic meshing tool was used to discretized the RVE in to tetrahedral elements for computing the results for the AEHM. Nodes on surface boundaries of the RVE can be distinguished in: corners, edges and faces. Furthermore, nodes can be distinguished in master nodes and slave nodes. Corners, edges and face which are slave nodes must be associated to master nodes

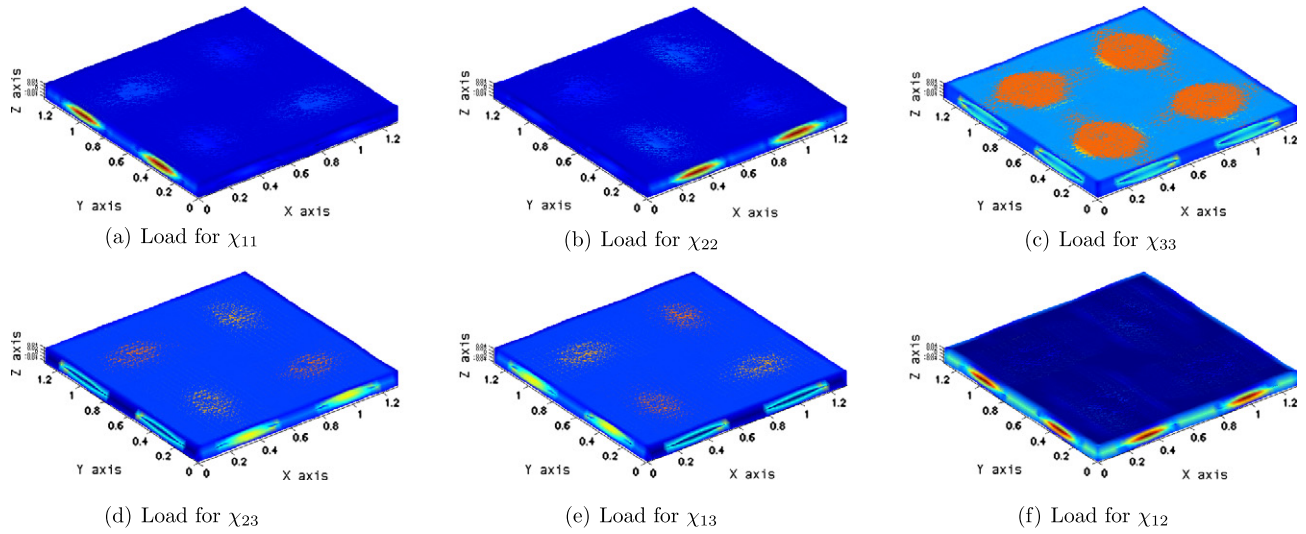


Fig. 5. Characteristic loading vectors for a RVE of plain weave fabric with gaps.

Table 1

Elastic properties of fibers and matrix [21].

Material	E_1 [GPa]	E_2 [GPa]	G_{12} [GPa]	G_{23} [GPa]	ν_{12} [-]
E-Glass	70.0	70.0	28.45	28.45	0.23
PPS	3.8	3.8	1.39	1.39	0.37

Table 2

Geometrical parameters [6].

a_f [mm]	h_f [mm]	g_f [mm]	a_f [mm]	h_f [mm]	g_f [mm]	V_f^c [-]	V_f^s [-]
0.95	0.11	–	1.04	0.11	–	0.50	–

Table 3

Results from models compared against experimental ones for E-Glass/PPS plain weave fabric.

	EXP [6]	AEHM	PP	PS	SP	SS	PP [6]
E_1 [GPa]	24	27.56	27.23	26.85	26.38	23.53	25.0
E_2 [GPa]	24	27.99	27.14	24.51	24.18	23.41	–
E_3 [GPa]	–	23.01	18.08	17.43	17.34	16.53	12.0
G_{12} [GPa]	4.7	6.03	5.87	5.68	5.67	5.44	5.8
G_{13} [GPa]	–	8.90	6.44	6.24	6.12	5.89	6.8
ν_{12} [-]	0.12	0.18	0.17	0.19	0.19	0.19	0.14
ν_{13} [-]	–	0.30	0.32	0.32	0.32	0.33	0.38

Table 4

Elastic properties of fibers and matrix [18].

Material	E_1 [GPa]	E_2 [GPa]	G_{12} [GPa]	G_{23} [GPa]	ν_{12} [-]
T-300 Carbon	231.0	40.0	24.0	14.4	0.26
PPS	3.8	3.8	1.39	1.39	0.37
PEI	3.0	3.0	1.1	1.1	0.36

Table 5

Geometrical parameters [18].

a_f [mm]	h_f [mm]	g_f [mm]	a_f [mm]	h_f [mm]	g_f [mm]	V_f^c [-]	V_f^s [-]
1.5	0.15	–	1.5	0.15	–	0.60	–

Table 6

Results from models compared against experimental ones for T-300 carbon/PPS 5H-satin-weave fabric.

	EXP [6]	AEHM	PP	PS	SP	SS	PP [6]
E_1 [GPa]	84	76.99	75.18	75.08	70.12	46.91	77.0
E_2 [GPa]	84	81.28	75.18	49.35	47.40	46.91	77.0
E_3 [GPa]	–	21.53	18.14	16.68	16.52	15.28	14.0
G_{12} [GPa]	4.1	5.51	5.42	5.22	5.20	4.96	5.9
G_{23} [GPa]	–	5.71	4.42	4.07	3.87	3.83	5.9
G_{13} [GPa]	–	5.74	4.42	4.41	4.13	3.83	5.9
ν_{12} [-]	0.02	0.01	0.02	0.03	0.04	0.04	0.05
ν_{13} [-]	–	0.57	0.56	0.55	0.54	0.59	0.44
ν_{23} [-]	–	0.57	0.56	0.61	0.61	0.59	0.44

Table 7

Elastic properties of fibers and matrix [27].

Material	E_1 [GPa]	E_2 [GPa]	G_{12} [GPa]	G_{23} [GPa]	ν_{12} [-]
E-Glass/Epoxy	55.7	18.5	6.89	6.04	0.22
E-Glass/Vinylester	57.5	18.8	7.44	7.26	0.29

on parallel faces. Fig. 3 displays the tetrahedral mesh for a RVE of a plain weave fabric with gaps, in particular for strands, matrix and complete RVE. As an unstructured mesh was used in this work there is not a one-to-one association between master and slave nodes. For this reason the code is programmed to find the best matching master node for each slave node, as discussed in [31]. Moreover, in order to prevent rigid body motion, the correctors of at least one arbitrary point in the RVE must be fixed. Two different methods were implemented for imposing the periodic boundary conditions: *Lagrange multipliers* and *static condensation* (or simply *condensation*). As only very small differences were found between the two methods, as condensation is more efficient, this method was used for calculating all the results for the AEHM.

Figs. 4 and 5 display, respectively, the microcorrectors and the characteristic loading vectors for a RVE of plain weave fabric with gaps.

3.1. E-Glass/PPS plain weave fabric

Akkerman [6] reported experimental results for an E-Glass/PPS plain weave fabric with elastic properties of fibers and matrix in

Table 8
Geometrical parameters [27].

a_f [mm]	h_f [mm]	g_f [mm]	a_f [mm]	h_f [mm]	g_f [mm]	V_f^c [-]	V_f^s [-]
0.6	0.09	–	0.6	0.09	–	0.52	0.80

Table 9
Results from models compared against experimental ones for E-glass/epoxy 8H-satin-weave fabric.

	EXP [27]	AEHM	PP	PS	SP	SS	PP [27]
E_1 [GPa]	25.6 ± 0.2	27.85	27.52	27.15	26.83	23.13	26.03
E_2 [GPa]	25.6 ± 0.2	28.50	27.52	24.13	23.87	23.13	26.03
E_3 [GPa]	–	16.70	21.48	15.98	15.92	15.05	15.65
G_{12} [GPa]	5.7 ± 0.3	6.21	5.94	5.69	5.67	5.36	5.67
G_{23} [GPa]	–	5.68	7.84	5.40	5.35	5.12	5.42
G_{13} [GPa]	–	5.68	7.89	5.52	5.45	5.12	5.42
ν_{12} [-]	0.13 ± 0.005	0.14	0.13	0.15	0.15	0.15	0.12
ν_{13} [-]	–	0.30	0.28	0.30	0.30	0.32	0.28

Table 1 and geometrical parameters in Table 2. No gaps were reported between the yarns. Given that the geometrical parameters of the fill and warp yarns are not identical, this is an unbalanced fabric. This fabric is non-hybrid.

In Table 3, the experimental results are compared against those predicted by the AEHM, the analytical models and other theoretical results for the same case found in literature. Note that the shape functions used by [6] are different from those used in the present study.

The AEHM gives similar results to the PP method, which is the best of the analytical methods.

3.2. T-300 Carbon/PPS 5H-satin-weave fabric

Akkerman [6] reported experimental results for a T-300 carbon/PPS 5H-satin-weave fabric with elastic properties of fibers and matrix in Table 4 and geometrical parameters in Table 5. No gaps were reported between the yarns. Given that the geometrical parameters of the fill and warp yarns are identical, this is a balanced fabric. This fabric is non-hybrid.

In Table 6 the experimental results are compared against those predicted by the AEHM, the analytical models and other theoretical results for the same case found in literature. Note that the shape functions used by [6] are different from those used in the present study.

Again, the AEHM gives similar results to the PP scheme, which is the best of the analytical methods.

3.3. E-Glass/epoxy 8H-satin-weave fabric

Scida et al. [27] reported experimental results for an E-glass/epoxy 8H-satin-weave fabric with elastic properties of fibers and matrix in Table 7 and geometrical parameters in Table 8. No gaps were reported between the yarns. Given that the geometrical parameters of the fill and warp yarns are identical, this is a balanced fabric. This fabric is non-hybrid.

In Table 9 the experimental results are compared against those predicted by the analytical model and other theoretical results for the same case found in literature.

The AEHM and the PP scheme yield similar results.

3.4. E-Glass/Epoxy plain weave fabric

Naik and Gamesh [25] reported experimental results for an E-Glass/Epoxy plain weave fabric with elastic properties of fibers

Table 10
Elastic properties of fibers and matrix [25].

Material	E_1 [GPa]	E_2 [GPa]	G_{12} [GPa]	G_{23} [GPa]	ν_{12} [-]
E-Glass	72.0	72.0	27.7	27.7	0.30
Epoxy	3.5	3.5	1.3	1.3	0.35

Table 11
Geometrical parameters [25].

a_f [mm]	h_f [mm]	g_f [mm]	a_f [mm]	h_f [mm]	g_f [mm]	V_f^c [-]	V_f^s [-]
0.86	0.11	0.00	0.84	0.11	0.02	0.46	0.74

Table 12
Results from models compared against experimental ones for E-Glass/Epoxy plain weave fabric N.3.

	EXP [25]	AEHM	PP	PS	SP	SS	SP [25]
E_1 [GPa]	22.8 ± 1.2	24.26	24.43	23.97	23.19	20.14	22.0
E_2 [GPa]	22.8 ± 1.2	25.21	24.19	20.84	20.42	19.65	–
G_{12} [GPa]	5.5 ± 0.1	4.75	4.69	4.51	4.50	4.31	4.55

and matrix in Table 10 and geometrical parameters in Table 11. This is another example of fabric with gaps between the yarns. Given that the geometrical parameters of the fill and warp yarns are not identical, this is an unbalanced fabric. This fabric is non-hybrid.

In Table 12 the experimental results are compared against those predicted by the analytical model developed in this work and other theoretical results for the same case found in literature. Only the in-plane shear modulus measured using ±45° off-axis tension tests have been reported. Note that the shape functions used by [25] are different from those used in the present study.

Here the AEHM and the PP scheme yield the best results. The predicted values are particularly good because the presence of gaps has been accounted for in the model.

4. Conclusions

This paper has assessed the capability of the asymptotic expansion homogenization method (AEHM) in comparison to analytical methods based on the mechanics of materials approach for predicting the mechanical properties of 2-D woven composites. In terms of analytical methods four types of averaging schemes are taken into account: the parallel–parallel, parallel–series, the series–parallel and the series–series schemes. Different types of weave patterns, in particular plain, five-harness satin and eight-harness satin weave, as well as different material properties were taken into consideration. Also unbalanced weaves and weaves with gaps were analyzed. The results are compared to experimental and numerical results for various cases found in literature.

The AEHM provides a good estimation of the mechanical properties for all the cases especially for the in-plane properties, while the out-of-plane properties are estimated less precisely, because of the assumption that the individual laminae are modeled as orthotropic materials. The AEHM compares well with the best results of the analytical methods which were provided by the parallel–parallel scheme. For all considered methods, the precision in the results is greatly improved when the geometry takes in to account the gaps between the yarns. Moreover, although not discussed here, the AEHM is able to provide localization of stresses and strains directly as part of the methodology [15,30,31], feature that is not provided by the analytical methods.

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